

# CS-IRCFS Platform

**Citizen Science for Irrigation Resilience and Climate-smart Food Security.** Farmers and monitors report from any phone through **USSD \*801#** and **SMS 8448**. District Planners then use those reports to check whether the **PADAB** and **APEFA Solar** irrigation schemes deliver their expected outcomes, and where they fall short.

Status as of 7 October 2026 · Work period 3 September – 6 October 2026 · Repository: cs-ircfs-platform

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architecture modules  
implemented end to end

**58**

API endpoints (FastAPI,  
documented at /docs)

**18 / 18**

automated tests passing  
(run 7 Oct 2026)

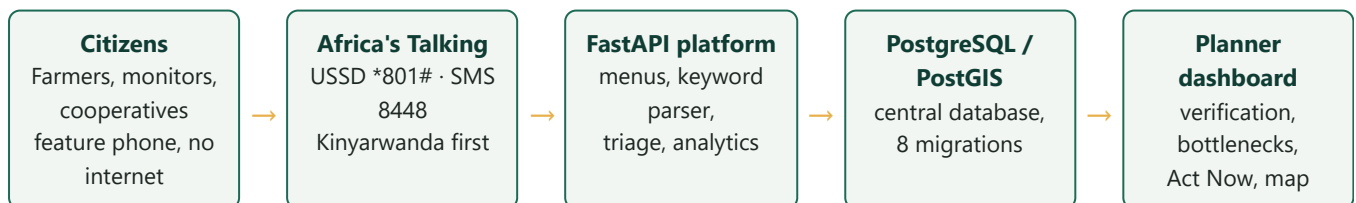
**15 · 72**

Bugesera sectors and cells  
loaded with boundaries

## 1. In short

- **The complete chain works:** phone → gateway → database → planner dashboard → SMS back to citizens. It can be shown live with the on-screen phone simulator.
- **All three modules** from the architecture document are built: food security, irrigation and climate resilience, and community feedback loops. The participation mechanics in §8 are built too: airtime rewards, local tips, Data Champions, and Kinyarwanda-first menus.
- **Outcome verification works:** farmer-reported harvests are compared with each scheme's yield target, and recurring bottlenecks are flagged automatically.
- **Ready to hand over:** Docker Compose with PostGIS, a hosted demo on Render, a one-command local demo, and six guides including a 10-minute presentation script.
- **Still needed before field use:** Africa's Talking production credentials and a short code. The sandbox already works. We also need the official PADAB and APEFA yield targets.

## 2. How it works



↩ The platform sends SMS back to citizens: local tips, irrigation advice, airtime rewards, and closing-the-loop messages

### 3. What the platform does

| Module                               | Feature                         | How it is delivered                                                                                                                                       |
|--------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Citizen science and food security | Crowdsourced yield forecaster   | USSD option 1 or UMUSARURO . Expected and reported harvest are compared with the scheme target                                                            |
|                                      | SMS pest and disease alerts     | NZANA / INDWARA keywords. Pest heatmap. Severity 4–5 opens an Act Now case automatically                                                                  |
|                                      | Household nutrition tracker     | 3 USSD questions give a stunting-risk score of 1–5, mapped by cell                                                                                        |
| 2. Irrigation resilience and climate | Infrastructure health reporter  | USSD option 4. Asset list from the PADAB/APEFA inventory. Faults are classed as technical, social, institutional or environmental                         |
|                                      | Citizen rain-gauge network      | USSD option 3 or IMVURA . Rainfall and drought-warning map layer                                                                                          |
|                                      | Irrigation scheduling assistant | 7-day rainfall per sector becomes threshold advice, sent by SMS to cooperatives                                                                           |
| 3. Community feedback loops          | Anonymous grievance log         | USSD option 5 or IKIBAZO . Only the cell is stored, never the reporter. Includes resettlement and downstream impacts                                      |
|                                      | Closing-the-loop SMS            | After a case is resolved: preview, then SMS to everyone registered in the affected cell                                                                   |
| Participation (\$8)                  | Incentives and local insight    | 100 RWF airtime for every 3rd weather or infrastructure report. A local tip after every report. Cooperative Data Champion roles. Numeric menus, ≤3 levels |
| Outcome verification                 | Objectives 1 and 2              | Scheme performance against verified yield targets (a target cannot be saved without its source). A bottleneck that recurs 3+ times in 90 days is flagged  |

### 4. Timeline

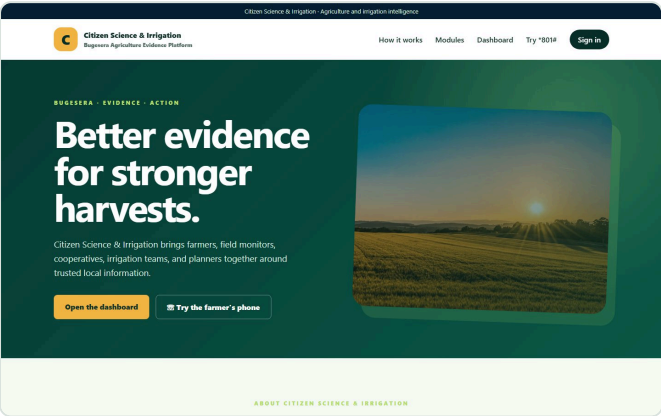
|           |                                                                                                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 Sep     | Architecture document ( <i>cs-ircfs-platform-architecture.docx</i> ) completed. This was the baseline for the build.                                                                     |
| 11–14 Sep | Initial schema, FastAPI platform, PostGIS geography, incident cases, and the Render hosted demo.                                                                                         |
| 15–22 Sep | Management filters and pagination, official Bugesera boundary on the map, VS Code launcher, and scheme sector/cell selectors.                                                            |
| 29 Sep    | Web accounts and authentication, new landing pages, and dashboard fixes.                                                                                                                 |
| 5–6 Oct   | SMS advisory, Google sign-in, the full USSD/SMS field channels, outcome verification, nutrition tracker, closing-the-loop SMS, presentation tour, and Africa's Talking sandbox delivery. |

## 5. What we built

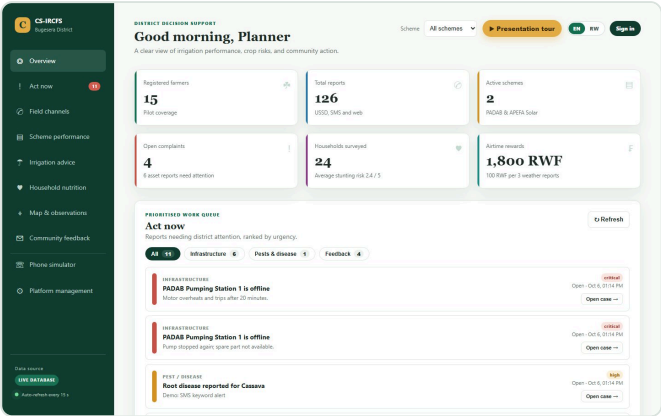
| Component                    | What is in place                                                                                                                                                                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Backend API</b>           | FastAPI service with 58 endpoints: channels (USSD/SMS), reports, cases, feedback, analytics, advisory, auth, management. Services cover USSD menus, SMS keywords, reporting, advisory, notifications and SMS/airtime delivery.                                                            |
| <b>Database</b>              | PostgreSQL + PostGIS with 16 tables: sectors, cells, schemes, users, crop logs, irrigation/climate logs, feedback with an audit trail, incident cases, nutrition surveys, inbound and outbound messages, rewards, accounts and sessions. All changes go through Alembic migrations 01–08. |
| <b>Field channels</b>        | Africa's Talking-format USSD and SMS callbacks. 6-option Kinyarwanda menu. Keyword parser. Every step is logged with the record it created. Sandbox delivery is working, and per-recipient SMS and airtime status is recorded.                                                            |
| <b>Planner dashboard</b>     | Overview, Act Now queue ranked by urgency, live field messages, scheme performance, response health, irrigation advice, nutrition risk, layered Leaflet map with the official Bugesera boundary, EN/RW switch, and a ► presentation tour.                                                 |
| <b>Phone simulator</b>       | A feature phone on screen that calls the real USSD/SMS endpoints, so the whole flow can be shown without a telecom contract.                                                                                                                                                              |
| <b>Platform management</b>   | Users, irrigation schemes (with verified yield targets), citizen reports, rainfall, feedback, nutrition surveys, SMS log, accounts and roles. Filters, pagination and CSV export.                                                                                                         |
| <b>Accounts and security</b> | Email sign-in (PBKDF2-SHA256, 210,000 rounds), hashed session tokens that expire after 12 hours, admin-only role promotion, and optional Google sign-in. Service API keys for integrations.                                                                                               |
| <b>Spatial data</b>          | 15 sectors and 72 cells loaded from GeoPackage (no QGIS needed). Official district boundary in GeoJSON.                                                                                                                                                                                   |
| <b>Deployment</b>            | One-command local demo ( <code>start-local.ps1 -Demo</code> ) on a separate database. Docker Compose with PostGIS for district handover. Render blueprint for a hosted preview. VS Code tasks.                                                                                            |
| <b>Documentation</b>         | Platform guide, architecture alignment (every architecture item mapped to code), data model, deployment, presentation demo script, and roadmap and limits.                                                                                                                                |
| <b>Quality</b>               | 18 automated tests (API, USSD, SMS, closing the loop, analytics, sign-in, SMS provider). They always run on an in-memory database, so they never touch real data.                                                                                                                         |

## 6. Screens

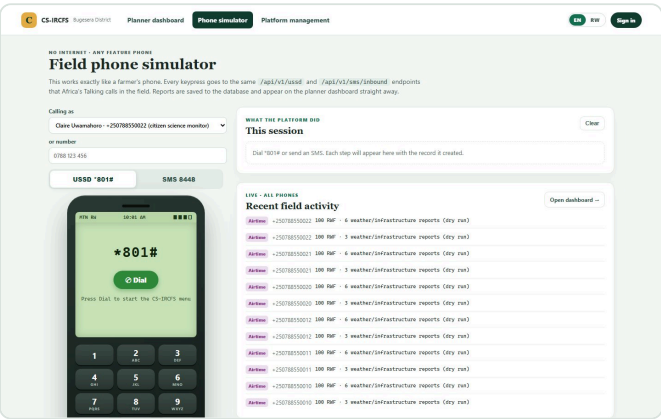
Captured on 7 October 2026 from the demonstration database. All records shown are invented demo data, and yield targets are placeholders.



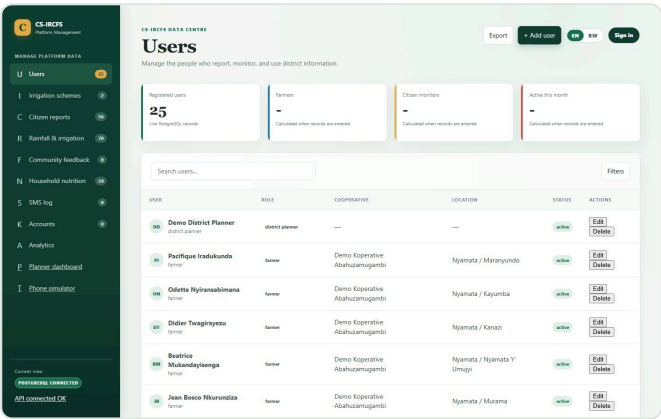
Home page: public introduction and how it works



Planner dashboard: overview figures and the Act Now queue



Phone simulator: dial \*801# or text 8448 against the real endpoints



Platform management: people, schemes, reports, accounts

### Example: outcome verification (demo data)

| Scheme      | Target (t) | Reported (t) | Achievement          | Infrastructure faults | Auto-flagged bottleneck  |
|-------------|------------|--------------|----------------------|-----------------------|--------------------------|
| PADAB       | 140.0*     | 28.3         | 20.2% · below target | 5                     | Technical (3 in 90 days) |
| APEFA Solar | 75.0*      | 55.4         | 73.9% · watch        | 1                     | None                     |

\* Placeholder targets until the official feasibility-study figures are entered.

## 7. Status against the six-month plan

| Phase                          | Deliverable                                                        | Status                                                                                                               |
|--------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Months 1–2: Design             | USSD wireframes, Kinyarwanda menu paths                            | <b>Done in software</b> 6-option menu, $\leq 3$ levels, numeric. Field validation with cooperatives is still needed. |
| Months 3–4: Development        | Backend, Africa's Talking, PostgreSQL, Leaflet dashboard           | <b>Done</b> Sandbox delivery works. Production needs approved credentials.                                           |
| Months 5–6: Pilot and handover | Pilot with 3 cooperatives, Docker, training and handover documents | <b>Ready to start</b> Docker Compose, operations guide and demo script are in place. The pilot itself is fieldwork.  |

## 8. Known limits

| Limit                 | What it means                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Live telecom delivery | Outside the sandbox, SMS and airtime are logged as <code>dry_run</code> until production credentials and a short code are approved.                  |
| Official baselines    | PADAB and APEFA yield targets are placeholders. Scheme performance shows "Target needed" until the real figures are entered.                         |
| Threshold calibration | The irrigation-advice thresholds (10 mm / 40 mm per 7 days) are starting values. They need calibrating from Mwesa Valley and Ngeruka/Mareba records. |
| Kinyarwanda review    | Phone screens and dashboard translations need review by native speakers.                                                                             |
| Area-level access     | Roles control what a user can do, but not which sectors they can see.                                                                                |
| Caller location       | A first-time caller has no cell until a planner adds one, so their reports are not mapped until then.                                                |
| Duplicate SMS         | An identical repeated SMS creates a second report. De-duplication by provider message ID is still needed.                                            |

## 9. Next steps

1. Get Africa's Talking production approval and a short code, then test on real phones with one cooperative.
2. Enter the official PADAB and APEFA yield targets, hectares and assets, each with its source.
3. Calibrate the dry- and wet-spell thresholds from historical rainfall records.
4. Have native speakers review the Kinyarwanda, and validate the menus with cooperative leaders.
5. Train 3 cooperative Data Champions, first on the simulator and then on their own phones.
6. Run the 3-cooperative pilot and export monthly CSVs for the pilot report.
7. Before scaling up: add a "choose your sector" first-call step, SMS de-duplication and area-level access. Hand over using `infrastructure/` and the guides.

**Before real citizen data:** production mode with strong secrets, HTTPS in front of the callbacks, daily off-server backups with a tested restore, a consent message for first-time callers, and a data-retention policy agreed with the district.